GHERBI Ikram

Data Engineer
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Profile

Data Engineer specialized in batch and streaming pipelines on a modern Big Data stack (Apache Spark, Apache Kafka, Apache Hudi, Apache NiFi, Trino, Docker). Designed and deployed an end-to-end Data Lakehouse architecture at Djezzy (Optimum Telecom Algérie) for real-time reporting of the 5G rollout, reducing reporting latency from 12–24h to real-time.

Technical Skills

Big Data & Data Processing: Apache Spark, Apache Kafka, Apache NiFi, Apache Hudi, HDFS, Apache Hive, Spark Structured Streaming

Storage & Databases: PostgreSQL, Oracle, MongoDB

Querying & Business Intelligence: Trino, SQL, Apache Superset, Power BI

Programming Languages: SQL, Java, C

DevOps & Tools: Docker, Docker Compose, Git, GitHub, VS Code

Data Modeling: UML, Merise, Medallion Architecture (Bronze/Silver/Gold), Data Lakehouse

Professional Experience

Data Engineer — End-of-Studies Internship (Big Data & Lakehouse Architecture) *February–June 2026*
Djezzy — Optimum Telecom Algérie, Algiers *Supervised by a Senior Big Data Architect*

- **Data Lakehouse Architecture (Bronze/Silver/Gold):** designed and implemented a real-time reporting solution for Djezzy's 5G rollout, reducing reporting latency from 12–24h to real-time. containerized infrastructure with Docker Compose.
- **Batch pipeline:** NiFi → HDFS → Hive → Spark/Scala → Apache Hudi MergeOnRead → Trino → Superset.
  computation of daily KPIs on 5G subscribers with multi-source joins.
- **Streaming pipeline:** Kafka → Spark Structured Streaming → Apache Hudi MergeOnRead → Trino → Superset.
  real-time updates of a central subscriber table with upsert by MSISDN.
- **Production-ready:** compiled Spark JARs for standalone execution, Silver tables exposed via Trino (Hive EXTERNAL TABLES), fully replayable pipeline after restart.

Computer Science Curriculum Writer *2025*
Private computer science school, Algeria

- **Curriculum design:** developed 3 structured training programs — (1) Computer science fundamentals (algorithmic thinking, Word, Excel), (2) Databases & web development (simple web project), (3) Advanced web development — totaling 32 structured sessions with learning objectives, practical exercises, and complete teaching materials.

Data Engineer — End-of-Studies Internship (Data Pipeline & Analytics) *February–June 2024*

Djezzy — Optimum Telecom Algérie, Algiers *Supervised by a Senior Big Data Architect*

- **NMS (Network Monitoring System) Pipeline:** ingested Djezzy network alarm data via Apache NiFi (3 separate CSV sources) with Avro Registry schemas, cleaning (header removal, invalid '?' characters), SQL deduplication, and loading into a PostgreSQL database — covering 50+ attributes per alarm (type, severity, ACK status, equipment IP, geographic location).
- **Power BI Network Monitoring Dashboard:** built an analytical dashboard featuring (1) a stacked histogram of alarms by type (10 categories: Trunk system, Communication system, Hardware, Environment, Running, Internal, QoS, Signaling, Processing error, Software) and severity (Critical/Major/Minor/Warning), (2) an interactive map of network equipment across Algeria (GPS coordinates, altitude/longitude).

Education

Master 2 — Advanced Information Systems Engineering *2025–2026*

*Hassiba Benbouali University, Chlef — **Thesis defense: 18.5/20***

Master 1 — Advanced Information Systems Engineering *2024–2025*

*Hassiba Benbouali University, Chlef — **Top of class***

Bachelor's Degree — Computer Systems *2021–2024*

Hassiba Benbouali University, Chlef

Baccalaureate, Experimental Sciences — 16.48/20 *2021*

Lycée Wancharissi

Languages

Arabic: Native | **French:** B2 (TCF) | **English:** C2 | **German / Korean:** Beginner

Interests

Reading | Artistic drawing